Senior High School

DEPARTMENT OF EDUCATION

Lesson Exemplar in General Mathematics

Quarter 1

Unit 3

Lesson Exemplar for General Mathematics
Quarter 1: Unit 1

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Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR					
Learning Area	General Mathematics			**Grade Level**	11
Semester	First	**Quarter**	First	**Unit**	3

I. OBJECTIVES	
Content Standard	*The learners demonstrate knowledge and understanding of financial application of sequence and series*
Performance Standard	*By the end of the quarter, the learners will be able to conduct a case study on payrolls for computing salaries, wages, deductions, and income for different employee profiles. Learners describe and present how they use sequences and series to make informed decisions on real-life situations involving earning money, buying, and selling, loans, investments, and purchases.*
Learning Competencies	The learners… 1. apply sequences and series to solve financial problems such as periodic payments for loans and mortgage. 2. determine the accumulated values of investment over time using sequences and series.
II. REFERENCES and MATERIALS	**Textbooks** Kirk Donna, 2023, *Contemporary Mathematics*, Open Stax, Rice University, Houston Texas Verzosa, Debbie Marie B. et al., 2016, *General Mathematics*, Commission on Higher Education, DepEd, Philippines

	Video lessons KhanAcademy:https://www.khanacademy.org/math/grade-8-math-tx/x42e41b058fcf4059:one-variable-equations-inequalities-and/x42e41b058fcf4059:simple-and-compound-interest/v/calculating-simple-compound-interest. Accessed 8 May 2025. **Application Software** • Kahoot • Plickers **Materials** • Calculators • Visual aids/PowerPoint presentation • Worksheets • Plickers cards • Budget template	
(These parts shall be accomplished per lesson)		
III. CONTENT	**Lesson 3. Applications of Sequences and Series in Financial Mathematics: Understanding Appreciation, Depreciation, Compound Interest, Loans, Mortgage Payment and Investment Growth** Recommended time allotment: 6 Hours	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. identify real-world financial applications involving sequences and series 2. determine the possible types of sequences and series that could be applied in financial problems and decision-making. 3. solve real-life financial problems by applying sequences and series	
V. PROCEDURES	**ACTIVITIES** (Instructions for Learners)	**ANNOTATION** (Instructions for Teachers)
A. Activating Prior Knowledge	**A.1. Activating Prior Knowledge** **Activity A.1: MY ONE MILLION JOURNEY** Instructions: Read and analyze the given scenario and answer the questions that follow. *Scenario*:	**Activity A.1: MY ONE MILLION JOURNEY** This activity aims to help learners recall their previous learning about sequences and series and apply these concepts to new financial problems. It will serve as a springboard for solving financial problems involving sequences and series. To facilitate this activity, follow these steps:

	Imagine it is 10 years from now. You have successfully graduated from college, started working, and built a good career. A bank is offering you a loan of ₱1 million. You now have an important decision to make. How would you use the money? You may choose one of the following options: • Buy a house • Buy a car • Start a business • Invest in life insurance *Processing Questions:* Ten years from the date you acquired your property (chosen from above), you have fully paid off your loan. Now, you want to sell your item to invest in something bigger (e.g., new business, real estate, expansion). • What would you do next? • How much do you think your asset is now worth? • Was your original decision a smart financial choice? Why or why not? • Do you think your property value would have increased or decreased after 10 years from the date you acquired your property? Why?	1. Begin by activating learners' prior knowledge. Ask learners to recall the concepts they learned about sequences and series. 2. Display pictures (such as a house, car, business, and life insurance) on the board, chart paper, or through a projector. 3. Facilitate the activity applying one of the suggested strategies: **Strategy 1: Think-Pair-Share** Have learners pair up and discuss the scenarios presented in the pictures. Invite pairs to share their opinions and reasoning with the class. **Strategy 2: Collaborative Group work** The learners will be randomly divided into five groups, with each group working together throughout the activity. They will discuss their ideas, share opinions, and collaborate in solving the problems presented. **Strategy 3: Peer to Peer Learning** Learners will choose their own group of five members based on their preference. Working closely as peers, each group will explore, discuss, and discover the processes needed to solve the given financial problems related to sequences and series. 4. Present a set of guided questions on the screen for learners to answer together. 5. Post or write the formulas for sequences and series on the board. Learners will use these formulas to solve financial problems 6. Guide learners as they perform calculations, allowing them to discover the process behind financial problem-solving. If possible, let them use tables for better understanding.

	A.2. Establishing the Purpose of the Lesson **Activity A.2: THE BEST INVESTMENT** Instructions: Analyze the given set of problems and decide which is a better investment, one that follows an arithmetic sequence or one that follows a geometric sequence? Why? Problem 1. Arithmetic Growth If the value of your chosen item increases by ₱50,000 every year, what would its value be after 10 years? Guide: Use Arithmetic Sequence Formula to find the value after 10 years Let a_n be the value of the property after n years. To find a_{10} the solutions are as follow: $a_n = a_1 + d(n-1)$ $a_{10} = 1050000 + 50000(10-1)$ a_{10}= 1,500,000 Year | Amount: 0 | 1000000; 1 | 1050000; 2 | 1100000; 3 | 1150000; 4 | 1200000; 5 | 1250000; 6 | 1300000; 7 | 1350000; 8 | 1400000; 9 | 1450000; 10 | 1500000 One good strategy to simplify calculations in problems of this kind is to adjust the units used for the variables. For example, we can let a_n = the value of the property in thousands of pesos after n years. Then $a_1 = 1050$ and $d = 50$. $a_{10} = 1050 + (10-1)(50) = 1050 + 450 = 1500$ Since this is in thousands of pesos, we multiply by 1000 to get the value of the property.	**Activity A.2: THE BEST INVESTMENT** This activity is designed to help learners analyze and compare two types of investment growth arithmetic and geometric by solving real-life problems involving the future value of a property over a 10-year period. Learners will use the arithmetic sequence formula to calculate value based on a fixed annual increase and the geometric sequence formula to determine value based on a percentage increase. The goal is to guide learners in understanding how different types of growth impact long-term returns and to help them develop mathematical reasoning skills for evaluating investment options. This activity also reinforces the practical application of sequences in financial decision-making. To facilitate the activity, you may apply any of the suggested strategies. **Strategy 1: Reflective Learning** Let the learners reflect and share the importance of the Lesson. Encourage them to share their realizations about the value of distinguishing between needs and wants, and how prioritizing essential needs over impulsive desires can lead to smarter financial decisions. **Strategy 2: Math Journaling** Learners will write their personal reflections, after which the teacher may randomly select and read some entries to highlight diverse perspectives Note *For the computations, learners should apply the formulas they have learned in sequences and series. Although formulas such as simple interest (A=Prt, etc.) are*

Nested table (Year/Amount) in the middle column:

Year	Amount
0	1000000
1	1050000
2	1100000
3	1150000
4	1200000
5	1250000
6	1300000
7	1350000
8	1400000
9	1450000
10	1500000

Problem 2. Geometric Growth
What if, instead of a fixed amount, your property's value increases by 5% each year? What would its value be after 10 years?

Year	Amount
0	1000000
1	1050000
2	1102500
3	1157625
4	1215506.25
5	1276281.56
6	1340095.64
7	1407100.42
8	1477455.44
9	1551328.22
10	1628894.63

Guide:
Let a_n be the value of the property in thousands of pesos after n years.

$$a_1 = 1000 \times 1.05 = 1050$$

The Geometric Sequence Formula is used because a common ratio (1+0.05) is multiplied to each term.
To find a_{10}, the solution is

$$a_n = a_1 r^{n-1}$$
$$a_{10} = 1050(1.05)^{10-1} = 1628.89463$$

The value of the property after 10 years will be ₱1628894.63

commonly used to calculate amounts, we will intentionally avoid them in this activity to emphasize the application of sequences and series in solving financial problems.

For Problem 2, explain why a geometric sequence is used. Since each term is multiplied by a common ratio of 1.05, it represents a 5% annual growth. (Be sure to explain that the ratio is 1.05 because 100% of the original amount plus an additional 5% growth equals 105%, or 1.05 in decimal form.)

B. Instituting New Knowledge	**B.1. Presenting Examples** **Activity B.1:** **UNDERSTANDING ESSENTIAL TERMS IN FINANCIAL MATHEMATICS** Instructions: Carefully study and understand the essential terms used in financial mathematics. These key concepts will help you interpret and solve real-life financial problems. **Essential Terms in Financial Mathematics** **Appreciation** is the increase in the value of an asset (like a house, land, or investment) over time. It often happens because of factors like demand, improvements, or inflation. Example: A lot owner bought a piece of land for ₱500,000. The value of the land appreciates by 8% every year. **Depreciation** is the decrease in the value of an asset over time. It usually happens due to usage, aging, or becoming outdated. Example: A car bought for ₱800,000 drops in value to ₱500,000 after several years. **Simple Interest-interest** that is computed on the principal and then added into it. **Compound Interest** - interest is computed on the principal and on the accumulated past interest Example: Saving ₱10,000 with 5% interest compounded annually means you earn interest not	**Activity B.1:** **UNDERSTANDING ESSENTIAL TERMS IN FINANCIAL MATHEMATICS** For learners to gain a better understanding of financial mathematics, key terms need to be clearly defined and thoroughly explained You may apply any of the following strategies. **Strategy 1: Fishbowl Discussion** 1. Prepare strips of paper with different terms written on them (Appreciation, depreciation, simple and compound interest, loan, mortgage, accumulated value of investment, conversion period, frequency of conversion, rate, etc.), place them inside a box. 2. Randomly pick a strip of paper from the box. 3. Using the same groupings, assign one member from each group to pick a term and provide their own explanation or definition of the term while the rest of the group listens attentively. 4. Repeat the process for each group, allowing different members to explain. 5. Assist learners who may struggle to define or explain their terms by providing guiding questions or hints. 6. Facilitate an interactive discussion, encouraging groups to ask questions, add examples, or clarify meanings to deepen their understanding.

	just on ₱10,000, but also on the interest you already earned. A **loan** is money borrowed from a bank or lender that you must pay back over time, usually with added interest. Example: Borrowing ₱1 million from the bank to buy a house and paying it back monthly with interest. A **mortgage** is a specific type of loan used to buy real estate (like a house or land). The property itself is used as a guarantee (collateral) until the loan is fully paid. Example: You borrow money from a bank to buy a house, and if you don't pay, the bank can take the house. **Accumulated value** (or future value) of an investment is the total amount an investment is worth after a certain time, including the original investment and all growth (appreciation or interest). Example: If you invest ₱100,000 today and it earns ₱50,000 interest in 5 years, the accumulated value is ₱150,000. **Conversion or Interest Period-time between successive conversions of interest.** Example: If compounded **annually** the conversion period is 1 year If compounded **quarterly** the conversion period is 3 months	**Strategy 2: Cabbage Paper Passing** 1. Prepare 5 paper "cabbages"by writing different terms on small pieces of paper. 2. Crumple each piece of paper into layers, forming a "cabbage" shape with multiple terms hidden inside. 3. Give one paper cabbage to each group. 4. While music plays, the group passes the cabbage paper around. 5. When the music stops, the learner holding the cabbage must peel off one layer (one piece of paper) and define the term they uncover. 6. Continue the process until all terms inside the cabbage are revealed and defined by different learners. 7. The teacher should assist learners who need help in explaining and encourage interactive discussion after each turn.

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<tr><td></td><td>If compounded monthly the conversion period is 1 month

If semi-annually the conversion period is 6 months

Frequency of conversion -the number of conversion periods in one year.

Example: If an account earns 5% annual interest compounded quarterly, the frequency of conversion is 4 times a year.

Rate – yearly rate of increase of the investment of the investment

Period Rate - the rate of interest for one conversion period

Time of term - how long in years the money is borrowed or invested.</td><td></td></tr>
<tr><td></td><td>B.2. Discussing the Concept

Activity B.2:
APPLICATION OF SEQUENCE AND SERIES IN FINANCIAL PROBLEMS

Instructions:
Financial problems, such as periodic payments for loans and mortgages can be solved using sequences and series. These mathematical tools are essential in determining the accumulated value of investments over time. Below are some examples that illustrate these applications. Participate in the discussion for better understanding,</td><td>Activity B.2:
APPLICATION OF SEQUENCE AND SERIES IN FINANCIAL PROBLEMS

This part of the lesson is crucial, as it involves facilitating an in-depth discussion of the solutions to each scenario with the class. As the teacher, you are expected to guide learners through the reasoning and steps involved in each solution, ensuring conceptual understanding. You may provide additional examples, clarify misconceptions, and apply any of the suggested discussion strategies to enrich the learning experience. Maximize the time so that the learners will gain understanding on this lesson.

Strategy 1: Real-World Problem Solving
1. Present a Realistic Scenario</td></tr>
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Example 1.

A lot owner bought a piece of land for ₱500,000. The value of the land appreciates by 8% every year. What will be the value of the land after 5 years?

STEP 1: Identify the variables

Let a_n = value of the land after n years

STEP 2: Identify what is being asked

We want to find the value of the land after 5 years, which is a_5

STEP 3: Identify the Given values

Given: a_1 = the value of the land after one year

$a_1 = 500000 \times 1.08 = 540{,}000$

$r = 1 + 0.08 = 1.08$

$n = 5$

STEP 4: Identify the series / sequence to be used:

Geometric Sequence : $a_5 = a_1 r^{5-1}$

STEP 5: Show the solution

$$a_5 = a_1 r^{5-1}$$
$$= 540000(1.08)^{5-1}$$
$$a_5 = ₱734664.04$$

STEP 6: Give the final Answer

The value of the land after 5 years is ₱734664.04

Example 2.

A cell phone was purchased for ₱32,500 in 2015. If the value of the cell phone depreciated annually by 16%. Find its value now. (2025)

2. Define the Problem Clearly
3. Gather and Analyze Relevant Information
4. Plan a Solution Strategy
5. Execute the Plan
6. Evaluate and Interpret the Solution
7. Reflect on Learning

Strategy 2: Scaffolding

1. Introduce the Problem
2. Activate Prior Knowledge
3. Break the Problem into Smaller Parts
4. Model the First Few Steps
5. Provide Guided Practice
6. Encourage Independent Problem Solving
7. Check and Discuss Solutions
8. Reflect and Summarize

STEP 1: Identify the variables
We measure the time in years beginning 2015. Let
a_n = the value of the cell phone n years after 2015

STEP 2: Identify what is being asked
We want the value of the cell phone in 2025 (ten years after 2015). Therefore, we want a_{10}.

STEP 3: Identify the Given values
Given: $r = 1 - 0.16 = 0.84$
$a_1 = 32500 \times 0.84 = 27300$
$n = 10$

STEP 4: Identify the series / sequence to be used:
Geometric Sequence : $a_{10} = a_1 r^{10-1}$

STEP 5: Show the solution
$$a_{10} = a_1 r^{10-1}$$
$$= 27300(0.84)^{10-1}$$
$$= \textbf{₱5684.29}$$

STEP 6: Give the final Answer
The value of the cell phone in 2025 is ₱5,684.29

Example 3.
₱3000 is placed in an account at 5% interest compounded quarterly for 5 years. How much is in the account at the end of 5 years?
Compound interest calculations take place every conversion period. Therefore, when using sequences, we take the number of conversion periods as our counter.

STEP 1: Identify the Variable
Let
a_n = the amount in the account after n conversion periods

STEP 2: Identify what is being asked
We want the value in the account at the end of 5 years. Since there are 4 conversion periods in a year, there will be $4 \times 5 = 20$ conversion periods in 5 years. We are therefore looking for a_{20}.

STEP 3: Identify the Given values
Given:
Yearly rate is 0.05. The rate per conversion period (per quarter) is $0.05 \div 4 = 0.0125$

$$r = 1 + 0.0125 = 1.0125$$

a_1 = the value after one conversion period (quarter)

$$a_1 = 3000 \times 1.0125 = 3{,}037.50$$

STEP 4: Identify the series / sequence to be used:
Geometric Sequence : $a_{20} = a_1 r^{20-1}$

STEP 5: Show the solution

$$a_{20} = a_1 r^{20-1}$$
$$= (3037.5)(1.0125)^{20-1}$$
$$a_{20} = \mathbf{3846.11}$$

STEP 6: Give the final Answer
the amount in the account after 5 years is ₱3846.11

Example 4.
Christina decides to save money for after graduation. Christina starts by setting aside ₱100. Each week, Christina increases the amount she saves by ₱20. How much money will Christina have saved after 52 weeks?

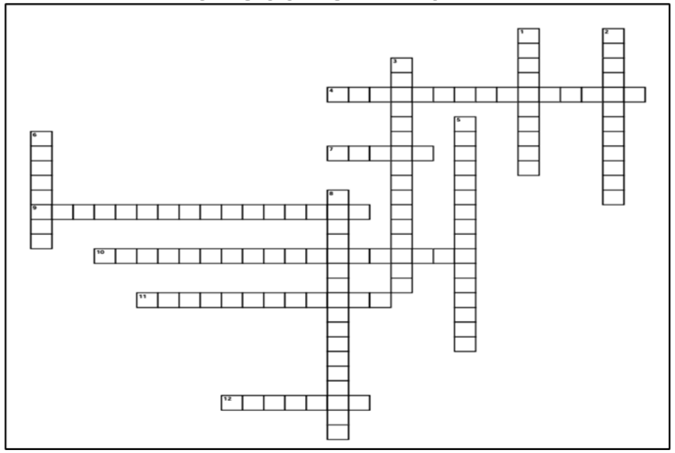
STEP 1: Identify the variable
Let
a_n = amount of money Christina will save in week n

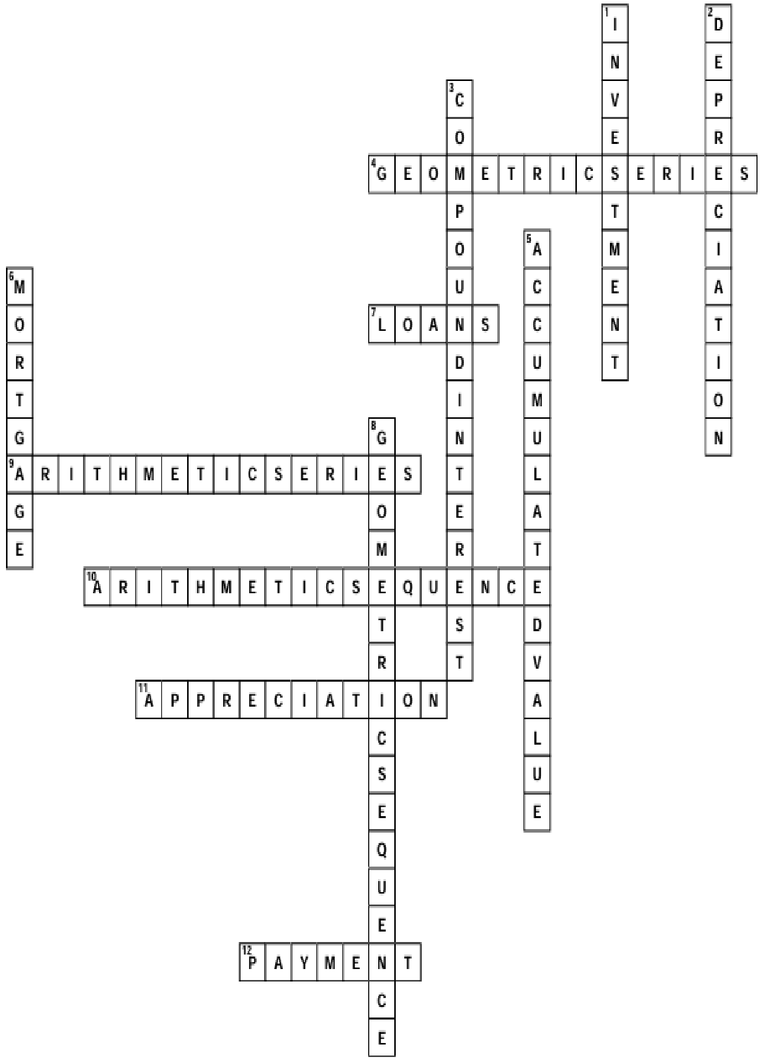
STEP 2: Identify what is being asked

	We want the total amount of money Christina will save for 52 weeks. What we want is $$S_{52} = a_1 + a_2 + \ldots + a_{52}$$ STEP 3: Identify the Given values Given: $a_1 = 100$ $n = 52$ $d = 20$ STEP 4: Identify the series / sequence to be used: Arithmetic Series Formula: $$S_n = \frac{n[2a_1 + (n-1)d]}{2}$$ STEP 5: Show the solution $$S_{52} = \frac{52[2(100) + (52-1)20]}{2}$$ $$= \frac{52(1220)}{2} = 31720$$ STEP 6: Give the final Answer The amount of money Christina will save 52 weeks is ₱31,720.00 *Example 5.* A savings account has an annual interest rate of 3% compounded monthly. Find the amount in the account 18 months after ₱5,000 is invested. STEP 1: Identify the variable Remembering that we count the sequence base on the number of conversion periods, we let	

	a_n = the amount in the account after n months STEP 2: Identify what is being asked We want the amount in the account 18 months after investing ₱5,000. This means we want a_{18}. STEP 3: Identify the Given values Given: The rate for one month is $0.03 \div 12 = 0.0025$ $r = 1 + 0.0025 = 1.0025$ $a_1 = 5000 \times 1.0025 = 5012.5$ STEP 4: Identify the series / sequence to be used: Geometric Sequence : $a_{18} = a_1 r^{18-1}$ STEP 5: Show the solution $a_{18} = a_1 r^{18-1}$ $a_{18} = 5012.5\,(1.0025)^{18-1}$ $a_{18} = 5229.85$ STEP 6: Give the final Answer The amount in the account after 18 months is ₱5,229.85	
	B.3. Developing Mastery **Activity B.3: PRACTICE PROBLEMS** Instructions: Solve the following: 1. A collection is taken up to support a family in need. The initial amount in the collection is P10,000. Everyone places P5000 in the collection. When the 35th person puts their P5000 in the collection, how much is present in the collection? 2. Ana, an investor, begins a start-up to revitalize Camela Homes. She begins with P1,000,000, making	Activity B.3: PRACTICE PROBLEMS These practice exercises are designed to reinforce learners' understanding of the concepts discussed in the lesson, particularly the application of arithmetic and geometric sequences in financial contexts. Through these activities, learners will independently solve problems that require the use of appropriate formulas and reasoning strategies. The exercises aim to strengthen computational skills, promote critical thinking, and allow learners to apply what they have learned to real-life financial scenarios. These also serve as formative assessment tools to help identify areas that may need further clarification or support.

	her investor number 1. Each investor that joins will invest P500,000 more than the previous investor. How much does the 10th investor invest in the project? With that 10th investor, what is the total amount invested in the project? 3. Alex and Jill deposit P4,000 in an account bearing 5% interest compounded yearly. If they do not deposit any more money in that account, how much will it be worth in 30 years? 4. A car was bought for P23,000. Its value is expected to depreciate by 2.5% annually. How much would the car cost after 3 years? 5. You buy a car through a ₱700,000 loan at 6% annual interest for 5 years. Meanwhile, the car depreciates by 10% each year. What is the car's value after 5 years? How much will you have paid in total for the loan by then?	To make the activities interactive you may used any of the following application software depends on the availability of the resources, **1. Kahoot Technology** Use Kahoot to create and input quiz questions. Learners can then respond interactively using their devices, making the activity more engaging and dynamic. **2. Plickers** Use Plickers to assess learners' understanding in real-time without the need for individual devices. Insert these questions in advance and have learners respond using printed cards, which you can scan with your mobile device.
C. Demonstrating Knowledge and Skills	***C.1. Finding Practical Application*** **Activity C.1: FINANCES THROUGH THE LENS** Instructions: Construct and present your own financial word problem that applies your understanding of sequences and series. The problem should be based on real-life financial situations such as compound interests, appreciation and depreciation, accumulated investment, loan and mortgage. You are encouraged to be creative in your presentation, you may use singing, dancing, acting, reporting, or any other expressive and engaging format that will help you clearly communicate your ideas.	**Activity C.1: FINANCES THROUGH THE LENS** This activity allows learners to apply their understanding of sequences and series in real-life financial contexts by creating and presenting their own word problems. By constructing original problems related to concepts such as compound interest, investment growth, depreciation, or loans, learners demonstrate both conceptual mastery and practical application of mathematical principles. The activity also encourages creativity and collaboration, as learners are given the freedom to present their work through expressive formats such as skits, songs, reports, or visual presentations. To facilitate this task, follow the procedures. Strategy: Experiential Collaborative Learning

	In your video, make sure to clearly state the financial problem, identify the type of sequence or series involved, explain the steps in solving the problem, and present the final answer with a brief interpretation.	1. Divide the class into 5 to 6 diverse groups, ensuring a mix of skills and backgrounds to promote inclusive participation and varied perspectives. 2. Provide a comprehensive overview of the activity, outlining the goals, expected outcomes, and how it integrates experiential learning principles. 3. Encourage learners to ask questions and provide thorough answers to ensure clarity and understanding of the task at hand. 4. Define specific deadlines for each phase of the project, including checkpoints for progress reviews, to help learners manage their time and stay on track. 5. After the completion of the activity, conduct sessions where learners can reflect on their experiences, discuss what they learned, and provide feedback on the collaborative process.
	C.2. Making Generalization **Activity C.2:** **REINFORCING KEY FINANCIAL TERMS** Instructions: Use the clues to fill in the crossword puzzle. Write your answers in the correct boxes for Across and Down. **CROSSWORD PUZZLE** 	**Activity C.2:** **REINFORCING KEY FINANCIAL TERMS** To conclude the lesson, inform the class that aside from using sequence and series formulas, there are also direct formulas for finding simple interest(A=Prt) and compound interest ($A = P(1+\frac{r}{n})^{nt}$). These formulas can be used to verify or cross-check their answers. Encourage learners to watch a video clip at home for additional learning: https://www.khanacademy.org/math/grade-8-math-tx/x42e41b058fcf4059:one-variable-equations-inequalities-and/x42e41b058fcf4059:simple-and-compound-interest/v/calculating-simple-compound-interest To summarize the unit lesson, a crossword puzzle has been prepared to review key concepts. The answer key is provided below for reference.

	ACROSS 4. Interest earned on past interest. 7. Borrowed money with interest. 9. Sum of an arithmetic sequence 10. Add the same number repeatedly. 11. Increase in asset value. 12 Money paid regularly for debt. **DOWN** 1. Money saved to earn profit. 2. Decrease in asset value. 3. Interest earned on past interest. 5. Principal plus earned interest. 6. Loan for buying property. 8. Multiply by constant ratio.	To end the lesson, you may inform the class that there is a formula for finding the simple interest. *Answer Key:* 

	C.3. Evaluating Learning **Activity C.3:** **ASSESSING THE SKILLS ON ARITHMETIC AND GEOMETICS SEUENCE** Instructions: Solve the following: 1. A farmer borrowed P30,000 at 8% interest compounded annually. How much will he have to pay at the end of 5 years? 2. A company's annual profit for the year 2022 was P210,000. The profit is expected to increase by 4% each year. Calculate the company's expected annual profit for the year 2027 and the accumulated profit from 2022-2027. 3. Liam starts a scholarship fund to support learners from rural areas. As the first donor, he donates ₱200,000. Each new donor contributes ₱50,000 more than the donor before them. How much does the 12th donor contribute to the fund? What is the total amount donated by the first 12 donors? 4. A community fundraiser begins with an initial donation of ₱15,000. Each supporter contributes a fixed amount of ₱3,000 to the fund. When the 40th person makes their contribution, how much money is now in the fund? 5. If P2,000 is deposited in an account that earns 8% interest compounded annually. Calculate the account balance after 4 years.	**Activity C.3:** **ASSESSING THE SKILLS ON ARITHMETIC AND GEOMETICS SEUENCE** This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation. You may provide additional items if necessary.

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C.4. Additional Activities

Activity C.4.1:
MORE ON SEQUENCES AND SERIES IN FINANCIAL MATHEMATICS

Instructions:
Solve each problem:

Remediation
1. Suppose P1000 is deposited by a 20-year-old worker in an Individual Retirement Account (IRA) that pays an annual interest rate of 12%. Describe the effect on the balance at age 65 if interest is compounded (a) annually and (b) quarterly.

2. A high school alumni group starts a donation drive with an initial fund of ₱20,000. Every batch that joins contributes ₱4,000. When the 25th batch contributes, how much money is in the fund in total?

3. A deposit of P2,500 is made in a savings account that pays an annual interest rate of 10%. Find the balance in the account at the end of 5 years if the interest is compounded
Quarterly
Monthly

4. An environmental group plants 100 trees in the first week. Every week after that, they plant 15 more trees than the previous week.

- How many trees are planted in the 10th week?
- How many trees have been planted in total after 10 weeks?

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C.4. Additional Activities

Activity C.4.1:
MORE ON SEQUENCES AND SERIES IN FINANCIAL MATHEMATICS

More problems have been provided in this part for more examples to give to learner.

This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

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5. A computer is bought for ₱50,000 and depreciates at 20% per year based on its current value. What will be its value after 3 years?

6. Suppose that P1000 is invested at an interest rate of 9% compounded monthly. Find the amount of investment (a) after 5 years, (b) after 10 years and (c) after 15 years.

7. Jared opens a tech startup with an initial capital of ₱500,000. Each new investor contributes ₱100,000 more than the previous one.

- How much does the 8th investor contribute?
- What is the total amount contributed by all 8 investors?

8. As part of a deal, a friend tells you they will give you P100 on day 1, P200 on day 2, P300 on day 3, for all 30 days of a month. At the end of that month, what is the total amount your friend has given you?

9. publishing company starts selling a new book with an opening day sale of 200 copies. Each following day, sales increase by 25 copies more than the previous day. How many books were sold on the 15th day?
What is the total number of books sold in the first 15 days?

10. A printer is bought for ₱15,000. It depreciates at 30% per year. What will be its value after 4 years?

Enhancement
1. Miles joined the Swimming Club in January 2025. The annual fee was P672. From 2025 the fees are expected to increase steadily by 4% every year

How much will the membership fee be in 2030?

	2. How much in total would Miles have paid in membership fees from 2025 to 2030? 3. A scholarship program offers a graduating Senior High School learner a starting annual allowance of P40,000, with a guaranteed 8% increase each year. • How much allowance will the learner receive in their 4th year? • How much in total allowance will the learner receive over the entire 4-year period? 4. The value of a certain book doubles every 10 years. In 1990, the book was worth P100. How much was it worth in 2000? In 2020? How much will it be worth in 2050? 5. A food bank distributes 120 food packs on the first day. Each day after, they increase distribution by 10 packs. How many packs are distributed on the 30th day? What is the total number of packs distributed over the 30 days? 6. A machine worth ₱1,000,000 depreciates by 15% each year. Find the value after 5 years.	
VI. ASSESSMENTS	Refer to the attachments.	
VII. REFLECTION	For the teachers, you may answer the following questions: 1. What went well in your teaching? 2. What challenges do you encounter? 3. How did you manage these challenges? 4. Were the learning objectives achieved by most learners? Why or why not?	

UNIT TEST

Direction: Choose the letter of the correct answer.

1. Which of the following best describes appreciation?
A. A steady decrease in asset value
B. A fixed amount deducted from an investment
C. An increase in asset value over time
D. Paying off a loan monthly

2. A car bought for ₱800,000 is now worth ₱500,000 after several years. What financial concept does this illustrate?
A. Appreciation
B. Loan Repayment
C. Compound Interest
D. Depreciation

3. Which scenario best illustrates compound interest?
A. Borrowing ₱1 million and paying it off monthly
B. Depositing ₱10,000 and earning interest on both the original amount and the interest gained
C. Buying a car that decreases in value
D. Saving ₱500 per month in a jar

4. Which of the following best describes a loan?
A. A gift from a bank
B. A one-time payment from your savings
C. Money borrowed and paid back with interest
D. The increase in land value

5. What makes a mortgage different from a regular loan?
A. It is paid back without interest
B. It requires no collateral
C. It is used specifically to buy property and is secured by that property
D. It cannot be paid monthly

6. What is the definition of a conversion period in finance?
A. The total time the money is invested
B. The time interval between each compounding of interest
C. The percentage added to the principal annually
D. The total amount of money in the account

7.What is the period rate in a compound interest problem?
A. The rate of interest for a conversion period
B. The amount of money deposited each time
C. The specific interval at which the interest rate is applied
D. The amount of interest earned at the end

8. An entrepreneur repays a loan by paying ₱10,000 in the first month, ₱9,000 the next, then ₱8,000, and so on. What type of sequence does this represent?
A. Geometric Sequence
B. Constant Sequence
C. Harmonic Sequence
D. Arithmetic Sequence

9. A car loses 15% of its value each year. What type of sequence does this represent?
A. Geometric Sequence
B. Arithmetic Sequence
C. Linear Sequence
D. Constant Sequence

10. Jamal deposits ₱2,000 every month in a bank. If he does this for 12 months, how much will he have saved?
A. ₱20,000
B. ₱22,000
C. ₱24,000
D. ₱26,000

11. What type of sequence/ series is the illustrated in item number 10
A. Arithmetic Sequence
B. Arithmetic Series
C. Geometric Sequence
D. Geometric Series

12. If ₱10,000 is invested at 6% simple interest annually for 3 years, what is the total amount at the end of 3 years?
A. ₱10,600
B. ₱11,610.16
C. ₱11,910.16
D. ₱11,800

For numbers 13-15. Use the problem below

If ₱5,000 is placed in an account at 6% annual interest compounded quarterly for 4 years, how much will be in the account at the end of the 4 years?

13. What is the period rate of the problem
A. 0.015 B. 1.015 C. 16 D. 5,075

14. What is the Conversion Period
A. 0.015 B. 1.015 C. 4 years D. 3 months

15. What is the answer to the question in the problem?
A. ₱6,344.93
B. ₱6,359.25
C. ₱6,374.95
D. ₱6,450.10

ANSWERS:
1.C 2.D 3.B 4.C 5.C 6.B 7.A 8.D 9.A 10.C 11. A 12.D 13.A 14.D 15. A

PERFORMANCE TASK: SAVING FOR A PURPOSE

Scenario:
Imagine you are saving money to achieve a personal goal, whether it's buying a gadget, planning a trip, or helping with family expenses. Each month, you set aside ₱500 into your piggy bank or savings account. (Feel free to adjust the monthly savings amount based on your goal. For an extra challenge, you can also factor in interest to see how your savings could grow even faster)

Task:
- Create a colorful and creative poster that tells the story of your savings over the course of one year.
- Design your timeline by creating a horizontal or vertical timeline that represents a full year, covering the months from January to December. Clearly label each month along the timeline to track your progress. This timeline will serve as a visual guide for planning and monitoring your savings or financial goals throughout the year.
- Track and illustrate your savings each month. Add ₱500 to your total savings for every month on your timeline. Beside each month, draw a simple illustration that shows your financial growth, such as a tree getting taller, a piggy bank filling up, or a staircase rising higher with each step. Along with your illustration, write down the total amount you have saved so far. For example, label January with ₱500, February with ₱1,000, March with ₱1,500, and continue this pattern for the rest of the year.
- Show your computation by using the sequence or series formula you learned in class to calculate your total savings.
- Finalize and decorate your poster by adding drawings or cutouts that represent your dream savings goals, such as a new phone, a vacation, a college scholarship, a small business, or a family car. Make your poster vibrant, colorful, and creative to visually showcase your aspirations and the hard work behind your savings plan.

Synthesis:
- How does saving small amounts regularly help reach big goals?
- What happens if you add interest (like banks do)?
- How did seeing the sequence make you feel about saving?

Rubric

Criteria	1	2	3	4	5	6
Mathematical Accuracy and Computation (6pts)	No computations shown or all are incorrect.	Formula used incorrectly or incomplete computation.	Used the correct formula but showed 3 or more computational errors.	Used the correct sequence but with 1–2 minor miscalculations	Minor error in computation or formula usage; final total is still correct.	Accurately used arithmetic/sequence formula with complete and correct computations for all months; final total is precise
	1	2	3	4	5	
Timeline Design and Organization (5pts)	Few months shown or unclear progression.	Timeline includes most months, but has noticeable disorganization.	Timeline shows all months but may be slightly hard to follow or not aligned.	Timeline is well-organized with minor labeling or spacing issues.	Timeline is clear, logical, and easy to follow with all months correctly labeled and in order.	
	1	2	3	4	5	6

Criteria	1	2	3	4	5	6
Monthly Visual Representation of Growth (6pts)	No illustrations shown or they do not reflect savings at all.	Few illustrations shown and not clearly connected to savings growth	Some illustrations are unclear or only partially related to savings.	Most illustrations are clear and relevant but may lack detail.	Illustrations show growth and match the savings per month with minor inconsistencies.	Illustrations for each month clearly show progressive financial growth with creativity and precision
Creativity and Visual Appeal (4pts)	1	2	3	4		
	Poster has minimal decoration or unclear representation of the dream.	Poster is neat with some decorative elements representing the dream.	Poster is visually appealing with good creativity and clear representation of dream.	Poster is vibrant, neat, highly engaging, and creatively showcases the learner's dream and journey.		
Representation of Dream Goal (4pts)	1	2	3	4		
	Dream goal is mentioned but not well-developed or visualized.	Dream goal is present and understandable.	Dream goal is clear and relevant but slightly disconnected from the savings timeline.	Dream goal is clearly illustrated and well-integrated into the savings story with personal meaning.		
Effort and Completeness (5pts)	1	2	3	4	5	
	Task is mostly incomplete or rushed.	Task is partially complete with some sections missing.	Task is mostly complete with effort shown in key areas.	Task is complete and well done; minor details may be missing.	Task shows exceptional effort and exceeds expectations; all components are complete.	